

JUSTIN W. SUNG

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EDUCATION

University of California, Berkeley

May 2025

Master of Engineering, Bioengineering (Bioinformatics & Computational Biology)

- GPA: 3.85
- Relevant Coursework: Computational Biology, Functional Genomics, Introduction to Machine Learning

University of Michigan - Ann Arbor

May 2024

Bachelor of Science, Molecular, Cellular, and Developmental Biology

- Relevant Coursework: Foundations of Bioinformatics and Systems Biology, Data-Oriented Programming, Algorithmic Thinking & Data Structures

EXPERIENCE

Lab Research Assistant | Merajver Lab

March 2023 – May 2024

University of Michigan Hospital Internal Medicine - Dept. of Oncology

Ann Arbor, MI

- Cultured and maintained TNBC (triple-negative breast cancer) cell lines for genomic and phenotypic assays
- Analyzed clinical transcriptomic data (RNA-seq, scATAC-seq) using custom R and Python scripts
- Ensured data quality through QC protocols and collaborated with interdisciplinary teams to resolve issues
- Maintained detailed lab documentation and contributed to improving experimental reliability

Emergency Medical Technician

May 2022 – Dec 2022

Emergent Health Partners - Huron Valley Ambulance

Ann Arbor, MI

- Delivered pre-hospital emergency care; assessed, triaged, and stabilized patients under time-critical conditions
- Documented patient data and clinical findings for hospital teams, emphasizing clarity and precision
- Gained deep exposure to real-world healthcare systems, improving contextual understanding of medical data

PROJECTS

Genetic Variant Predictor | UCSF Bakar Computational Health Sciences Institute

- Built a full-stack platform to automate scoring of genetic variants using protein language models and Ensembl's VEP
- Wrote modular Bash/Python backends to automate file ingestion, model inference, and structured output using APIs and batch pipelines
- Developed Flask APIs and React-based dashboards featuring progress indicators, tooltips, and error handling to deliver user-friendly data views
- Collaborated cross-functionally with UCSF clinicians and engineers, aligning model outputs to real-world diagnostic needs

SKILLS

Languages & Tools: SQL, Python, R, Bash, Git, Excel

Bioinformatics & Data Analysis: Bowtie, Samtools, RNA-Seq, ATAC-seq, VEP, ESM1b

Molecular Biology: Cell culture, DNA/RNA extraction, qPCR, PCR, gel electrophoresis